

ALPR System

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Image Processing project report

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INTRODUCTION TO ALPR SYSTEMS

Automated License Plate Recognition (ALPR), also known as Automatic Number Plate Recognition (ANPR), is a surveillance technology that utilizes Optical Character Recognition (OCR) on images to read vehicle registration plates. The system captures an image of a vehicle, locates the license plate within the frame, and extracts the alphanumeric characters into a digital text format.





THE PROCESS

The Problem

The primary challenge ALPR addresses is the "Identification Gap" in very high-speed situations where typical human observation is either inefficient or inaccurate.

The ALPR then is a major helping hand in these situations, as it provides an almost instantaneous identification for these specific areas.

Law Enforcement & Security

Manual checking of license plates against vehicles is very difficult for officers on a highway. ALPR allows for instantaneous scanning of thousands of plates per hour, informing authorities of vehicles of interest efficiently.

Other examples of the solutions

In Electronic Toll Collection (ETC) systems, ALPR eliminates the need for physical toll booths and human collectors, facilitating "Open Road Tolling" which significantly reduces traffic congestion and carbon emissions from idling vehicles.

Commercial facilities and gated communities use ALPR to automate entry and exit. This provides digital tracing of all vehicles available without requiring physical passes or staff to manage gates.

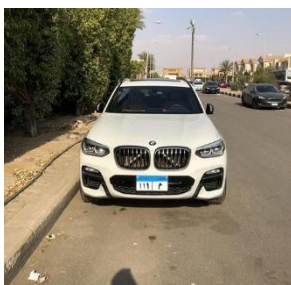
Dataset collection

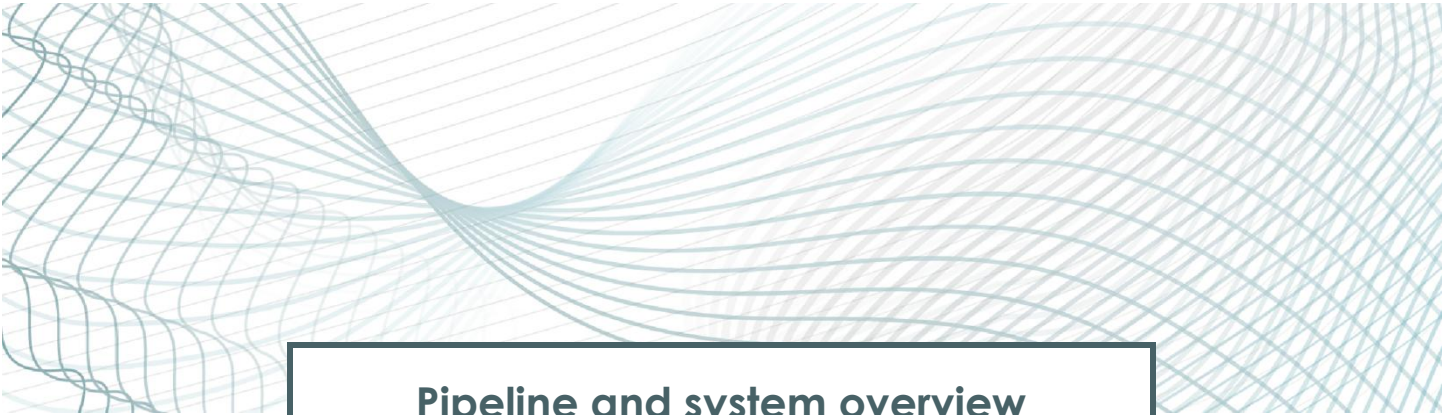
How it was collected

The dataset used in this project consists of real-world photographs of Egyptian civilian vehicles captured by team members using mobile phone cameras under varied environmental conditions.

All images were captured in JPEG format and stored in a dedicated dataset directory. The collection represents diverse real-world scenarios encountered in Egyptian urban and suburban environments

Examples





Pipeline and system overview

Methodology and stages

The ALPR pipeline is organized as a sequential series of processing stages, each solving a specific sub-problem and feeding its output to the next stage. Next figure provides a high-level overview of the complete pipeline. Every stage is implemented as either inline code within the main script (main_alpr.m) or as a modular helper function, enabling unit testing and independent debugging.

Figure description and roadmap

```
INPUT IMAGE (JPG)
|
v
[Stage 1] Image Standardization    --> Resize to 800px width
|
v
[Stage 2] Color Space Conversion   --> RGB to Grayscale + HSV
|
v
[Stage 3] Noise Filtering          --> Median Filter [3x3]
|
v
```

```

[Stage 4]  Color Segmentation          --> Blue Mask + White Mask
    |
    v
[Stage 5]  Sobel Edge Detection        --> Vertical + Horizontal (thr x 0.7)
    |
    v
[Stage 6]  Morphological Processing    --> Dilate [3x15] + Close + Fill
    |
    v
[Stage 7]  Connected Component Anal.  --> bwlabel + regionprops
    |
    v
[Stage 8]  Candidate Scoring          --> Multi-criteria score accumulation
    |          |
    |          [Fallback: Blue-band expansion if no candidate]
    |
    v
[Stage 9]  Plate Cropping             --> imcrop + 5px padding
    |
    v
[Stage 10] Adaptive Binarization      --> robust_binarize() dual polarity
    |
    v
[Stage 11] Character Segmentation      --> segment_characters()
    |
    v
[Stage 12] Zone Separation            --> Numbers (left) | Letters (right)
    |
    v
[Stage 13] HOG Feature Extraction     --> extractHOGFeatures [42x24, 8x8]
    |
    v
[Stage 14] Template Matching (OCR)    --> Euclidean distance, score>0.1
    |
    v
OUTPUT: Plate String + PNG + Excel

```

Stages demonstration

1. Image Standardization and Preprocessing

- Image Resizing:

All input images are resized to a fixed width of 800 pixels while maintaining the original aspect ratio.

This is accomplished via the MATLAB function `imresize(img_orig, 800/orig_w)`, which computes a scale factor as the ratio of 800 to the original image width and applies it uniformly to both dimensions. The 800-pixel target was selected to balance computational efficiency with sufficient spatial resolution.

At this resolution, a typical Egyptian plate (occupying roughly 5–15% of the vehicle frame width) subtends 40 to 120 pixels in width — sufficient for character edge detection and HOG feature extraction without excessive memory or computation overhead.

- Grayscale Conversion:

The resized RGB image is converted to a single-channel grayscale representation using **`gray_img = rgb2gray(img)`**. The MATLAB `rgb2gray` function applies the ITU-R BT.601 luminance equation:

$$L = 0.2989 R + 0.5870 G + 0.1140 B$$

This weighted average reflects the human visual system's greater sensitivity to green wavelengths and provides a perceptually meaningful intensity representation. The grayscale image is subsequently used for edge detection, adaptive thresholding, and HOG feature extraction.

- HSV Color Space Conversion:

In parallel with grayscale conversion, the RGB image is also transformed into the Hue-Saturation-Value (HSV) color space using **`hsv_img = rgb2hsv(img)`**. The HSV decomposition is defined as:

$$H = \text{hue angle (0-1 normalized)}, S = \text{saturation}, V = \text{brightness}$$

The HSV representation is preferred for color-based segmentation over RGB because it decouples chrominance (H, S) from luminance (V). Consequently, the hue of the Egyptian plate's blue strip remains approximately stable across different ambient illumination conditions, whereas its RGB values would vary substantially. This property makes HSV-based thresholding far more robust to lighting changes than RGB thresholding.

2. HSV Color Space Segmentation

- Blue Mask Generation:

A binary blue mask is generated by thresholding the three HSV channels simultaneously:

$$\text{blue_mask} = (H > 0.52 \text{ AND } H < 0.75) \text{ AND } (S > 0.25) \text{ AND } (V > 0.12)$$

In MATLAB's normalized HSV scale (0 to 1), the hue range 0.52–0.75 maps to approximately 187°–270° in standard angular representation, spanning cyan-blue through pure blue to blue-violet. This range encompasses the specific shade of blue mandated for Egyptian civilian plates. The saturation lower bound of 0.25 excludes achromatic regions (gray surfaces, roads, concrete) which have near-zero saturation. The value lower bound of 0.12 excludes near-black regions corresponding to night shadows or dark paint. After thresholding, the mask is morphologically closed with a 3×8 rectangular structuring element to bridge gaps within the strip, and small blobs (< 50 pixels) are removed via **bwareaopen** to eliminate noise reflections.

- White Mask Generation:

A white mask identifying high-brightness, low-saturation regions is generated as:

$$\text{white_mask} = (S < 0.20) \text{ AND } (V > 0.60)$$

White and near-white pixels have very low saturation (the color is washed out toward neutral) and high brightness. These thresholds capture the white plate background while excluding the darker road surface, vehicle body panels, and foreground objects. A minimum area filter of 100 pixels is applied to remove spurious white highlights from headlights or reflections

3. Plate Localization and Cropping

Once the optimal bounding box is determined, the plate region is extracted from the grayscale image using `imcrop` with a 5-pixel padding added to all four sides. This padding compensates for any under-estimation of the plate boundary by the scoring bounding box, ensuring that the full character shapes — including descenders and ascenders — are included in the cropped region. Boundary clamping (**max(1,...)** and **min(...,img_dims)**) prevents out-of-bounds cropping.

4. Character Segmentation

Character segmentation extracts individual character bounding boxes from the binary plate image. The **segment_characters()** function applies regionprops to the binary image to retrieve connected component bounding boxes and then filters them using a set of geometric constraints.

Parameter	Value	Rationale
MIN_WIDTH_RATIO	0.01	Character $\geq$ 1% of plate width
MAX_WIDTH_RATIO	0.30	Character $\leq$ 30% of plate width
MIN_HEIGHT_RATIO	0.10	Character $\geq$ 10% of plate height
MAX_HEIGHT_RATIO	0.85	Character $\leq$ 85% of plate height
ASPECT_RATIO_MIN	0.05	Not excessively narrow
ASPECT_RATIO_MAX	3.00	Not excessively wide
Top exclusion zone	Center Y < 30% of height	Exclude EGYPT/msr header
Edge exclusion zone	Center X < 3% or > 97%	Exclude plate frame artifacts
Central divider exclusion	Width < 1.5%, X in 48%–52%	Exclude zone separator line

After geometric filtering, detected character boxes are sorted by their x-coordinate (left-to-right). They are then spatially partitioned at 48% of the plate width into a numbers zone (left side) and a letters zone (right side), corresponding to the standard Egyptian plate layout. Each zone is matched against the appropriate character class in the OCR stage.

5. Adaptive Plate Binarization

Converting the grayscale plate crop to a binary image is a critical step that directly determines the quality of subsequent character segmentation. Illumination conditions dramatically affect whether characters appear as dark-on-white (standard daytime) or light-on-dark (night reflections, overexposed backgrounds). The `robust_binarize()` function addresses this challenge through a dual-polarity majority-vote approach.

This includes various methods such as:

- Contrast Enhancement
- Dual-Polarity Thresholding
- Polarity Selection

Results & Accuracy

Results table:

Filename	Is_Detected	Letters	Numbers
1.jpg	TRUE	م ط ر	٥ ٥ ٥
10.jpg	TRUE	ر س م	٧ ٨ ٩ ٧
11.jpg	TRUE	ر ص و	٤ ٩ ٦ ٧
12.jpg	TRUE	ر ع أ	٧ ٧ ٢ ٩
13.jpg	TRUE	ر س هـ	٢ ١ ٣ ١
14.jpg	TRUE	ر أ د	٢ ٩ ٥ ٣
15.jpg	TRUE	ر ب ط	٥ ٨ ٢ ٨
16.jpg	TRUE	م ن س	٦ ٦ ٦ ٦
17.jpg	TRUE	م ف	١ ١
18.jpg	TRUE	ب ط ل	٩ ٥ ٩
19.jpg	TRUE	ي ي ي	٩ ٩ ٩
2.jpg	TRUE	و و و	٦ ٦ ٦ ٦
20.jpg	TRUE	م ج د	٤ ٤ ٤
3.jpg	TRUE	ر ن أ	٨ ٦ ٤ ٨
4.jpg	TRUE	ر ف ي	٤ ٩ ٨ ٧
5.jpg	TRUE	أ ن ج	٢ ٣ ٥ ٩
6.jpg	TRUE	أ ف ط	٤ ٥ ٩ ٥
7.jpg	TRUE	ر س ن	٥ ١ ٥ ٩
8.jpg	TRUE	ر س أ	٢ ٨ ٩ ٥
9.jpg	TRUE	د ب ج	٩ ٨ ٨ ٧

Successful cases:



1. Standardized



2. Filtered Gray



3. Blue Mask



4. Edge Candidates



5. Detected (54.3)



6. Plate Bottom



7. Validated (7 chars)



8. Final Result

Result: DETECTED

EGYPT	مصر
٧ ٨ ٩ ٤	ر ف ي

1. Standardized



2. Filtered Gray



3. Blue Mask



4. Edge Candidates



5. Detected (51.7)



6. Plate Bottom



7. Validated (7 chars)

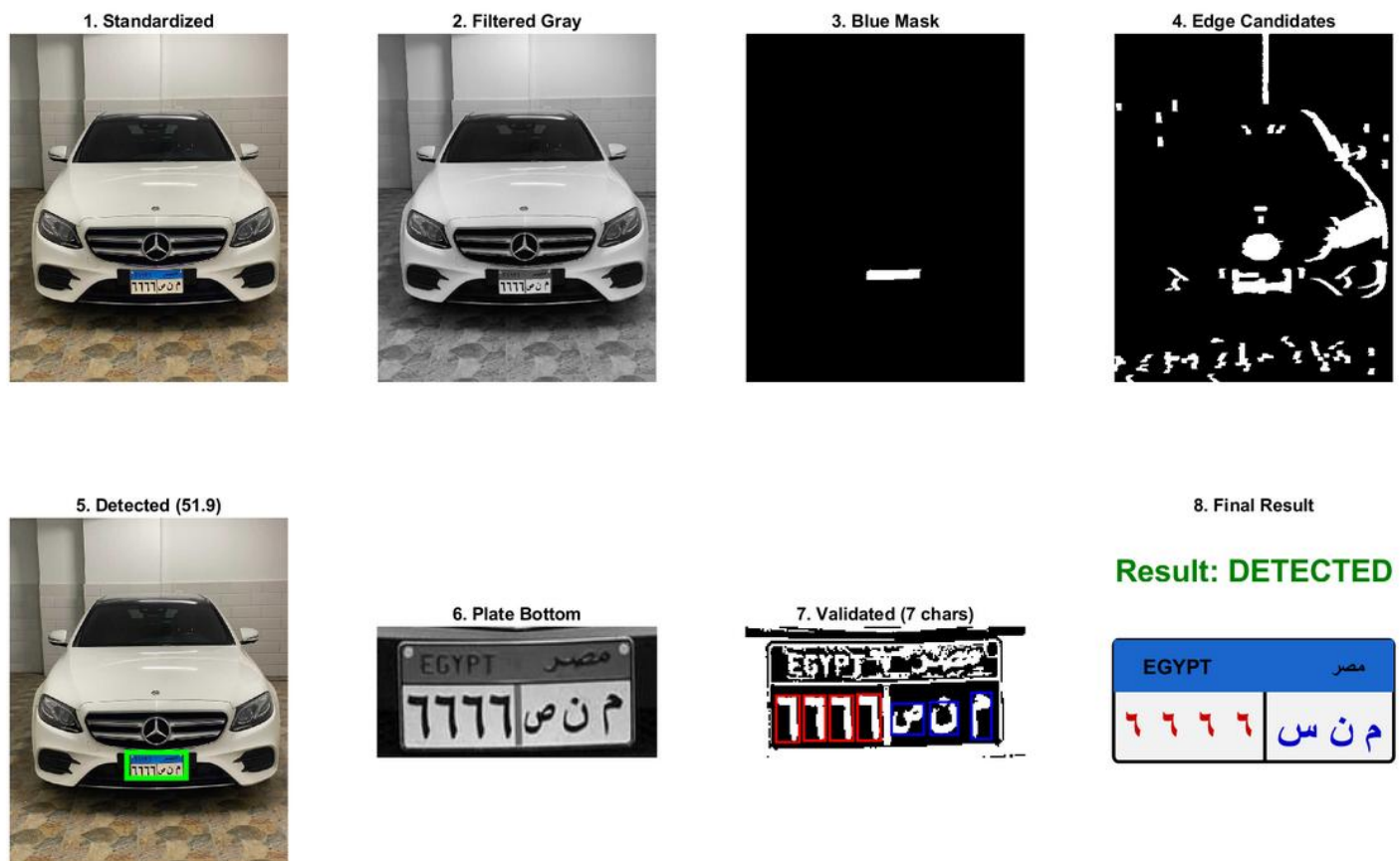
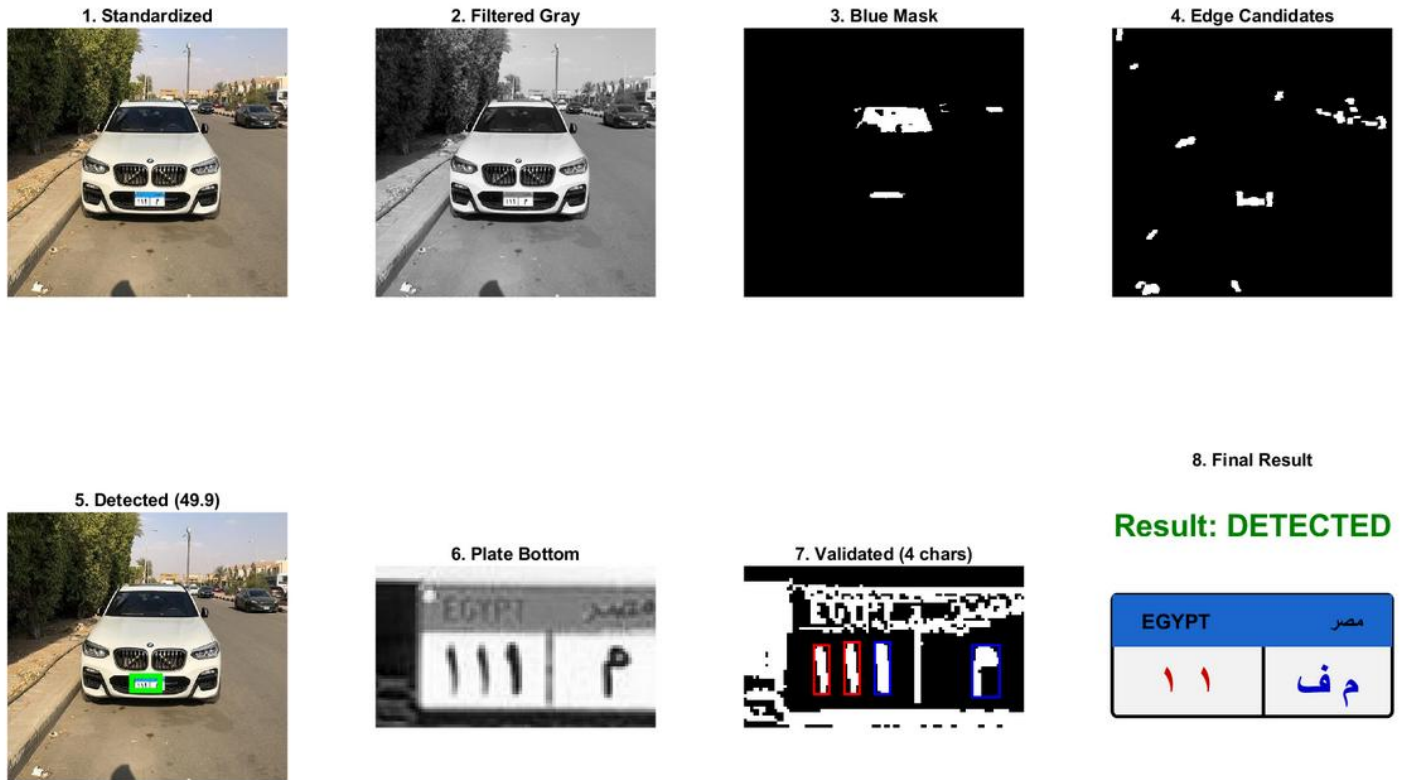


8. Final Result

Result: DETECTED

EGYPT	مصر
٩ ٥ ٣ ٢	أ ن ج

- Failed cases:



Discussion

- What worked well

The most significant engineering success of this project is the multi-criteria candidate scoring system.

By requiring convergent evidence from at least three independent sources — edge density, color composition, and geometric shape — the scoring function consistently selected the correct plate region even in cluttered backgrounds.

The dual-polarity adaptive binarization proved highly effective. Without this mechanism, images captured under artificial lighting (which can invert the apparent plate contrast) would produce near-empty binary images with no recognizable characters. The majority-vote design, requiring agreement from two out of three independent thresholding methods, provided excellent robustness against the failure of any single threshold computation.

The HOG-based OCR with mode-based filtering (number mode vs. letter mode) produced correct character identifications in the tested cases. The spatial separation at 48% of plate width reliably partitioned the digit zone from the letter zone, enabling the mode flag to eliminate cross-category confusion. The right-to-left output reversal via `flip()` correctly produced Arabic-readable plate strings.

- What failed

The most persistent challenge was interference in the HSV blue mask from non-plate blue objects in the scene.

Template coverage presented another significant challenge. The HOG matching quality is directly limited by the completeness of the template library. Characters whose templates have limited photographic variation (only one or two samples) are more susceptible to recognition errors when the character's on-plate appearance differs from the template due to illumination, scale, or wear.

Cases involving mild perspective angle (vehicles not photographed perfectly head-on) produced slight trapezoidal distortion of the plate, which shifts character proportions. Since the HOG features are not explicitly perspective-invariant, this can reduce similarity scores and occasionally produce character substitution errors.

- What could be improved

1. Short-Term Improvements

- Perspective rectification: Estimate the plate's four corner points using edge detection and apply a projective transform (homography) to de-skew angled plates before binarization, directly improving HOG feature quality.
- Connected character splitting: Implement a vertical projection histogram analysis to automatically split merged character blobs at local minima of the column pixel count profile.
- Template augmentation: Synthetically generate additional template variants by applying small rotations ($\pm 5^\circ$), Gaussian blur, and brightness variations to existing templates, increasing the effective template library size without manual photography.
- Adaptive zone separator: Dynamically estimate the digit/letter boundary from the detected character cluster centroids rather than using a fixed 48% threshold.

2. Medium-Term Improvements

- SVM classifier: Replace Euclidean HOG nearest-neighbor matching with a one-vs-all SVM trained on a large collection of segmented Egyptian plate characters. SVMs learn discriminative decision boundaries and are more robust to within-class variation than nearest-neighbor.
- Plate type classifier: Add a preliminary stage to classify the plate type (civilian, commercial, taxi, motorcycle) and switch detection and recognition parameters accordingly.
- Statistical language model: Incorporate a character transition probability model derived from valid Egyptian plate combinations to post-correct improbable character sequences.
- Video integration: Extend the pipeline to process video frames with temporal smoothing, tracking the plate bounding box across frames using a Kalman filter to reduce per-frame jitter.

3. Long-Term Research Directions

- End-to-end deep learning: Train a YOLOv8 or similar detection model on a large annotated Egyptian plate dataset for robust plate localization, coupled with a CRNN (Convolutional Recurrent Neural Network) with CTC loss for sequence-level OCR.
- Nighttime specialization: Train a dedicated image enhancement module using paired day/night images of the same plates to normalize appearance before detection.
- Multi-country support: Extend the template library and HSV color models to support additional Arabic-speaking country plate formats (Saudi Arabia, UAE, Jordan).

Work distribution

1. Ahmed Mohamed Attia Mohamed [2870]

- Group lead & Code implementation

2. Anas Ali Hamad Elsayed [2877]

- Report & presentation

3. Abdelhay Lotfy Elsayed [2894]

- Dataset collection

4. Ali Elshawadfy Abdallah [2899]

- Code implementation

5. Omar Hosni Mohammed Ahmed [2901]

- Report & presentation

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